

EXP1-2 Melon Bungee Evaluation Report P12

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Abstract: This report will show you our melon bungee result, analyze the difference between actual data and our prediction, and our reflection on this experiment.

Section1-Results:

Descending displacement: 4.1m / Actual rebound: 2.4m

For the descending part, bungee tape kept the melon safe while stretching its maximum length. Throughout the descent, there was no swaying or breaking of the rope. Although there was a long distance from the ground, the overall was very successful. ***For the rebound part, the tape also made the maximum contraction,*** bringing the melon at its lowest point back 40 percent of its descending displacement.

Section2-Comparison to prediction:

Predicted descent displacement: 4.178m / Error with prediction: 1.8%

Predicted rebound: 4.176m / Error with prediction: 42.5%

Our predictions are pretty good for the falling part, but the gap between our prediction of the rebound period and the actual data is large. The main reasons we analyzed for this error in the rebound part are as follows:

1. ***The elasticity of the tape decreased sharply after going through the falling stage,*** resulting in the stiffness of the tape in the rebound stage being far less than we predicted.
2. ***The tape's structure was complex, and the relationship between deformation distance Δx and force F couldn't be calculated simply by Hooke's theorem.*** Moreover, the unpredictable nature of the fall made it impossible to determine exactly how the rope changed shape, which also produced a large error in stiffness coefficient and rebound distance.
3. Because of the special structure of our tape, ***we could not see the tape as a simple spring,*** which meant not all gravitational potential energy would change into elastic energy. ***Also, we ignored the influence of air resistance.*** The lack of these situations led us to misjudge the height of the rebound.

Section3-Summary:

In this experiment, every member of the team had a deep understanding of their obligations and truly achieved the unity of the individual and team. In the beginning, everyone proposed their ideal bungee jumping scheme in the brainstorming stage. We discussed whether the scheme is feasible, and determined the most feasible one. Also, we identified the responsibilities of each member. Next, we got our equipment, like a spring dynamometer, elastic tapes, and hooks, and measured our data. When measuring stiffness, the strength of the team is perfectly reflected. Two or three people controlled the stretch band when one was reading and one was recording. Then, data processing was also the division of labor cooperation. Finally, it was time for the melon bungee jump. We sent two people upstairs to release the watermelon. Our safety officer protected all of us and other teammates took videos downstairs to record the experiment's completion. ***Each of us embodied our values, so the team worked very well in this experiment.***

Based on the summary and reflection of our experiment, we think that the most important thing we can do in the next experiment is pre-experiment. Before the formal experiment, we should find out the best experimental conditions to avoid the error caused by the experimental environment and equipment. In addition, we should have a more careful understanding of the scoring criteria when designing the experiment. For example, the best height of melon from the ground in this experiment should be less than 50cm. However, we were too conservative in designing the elastic rope binding method, which resulted in a low score. A pre-experiment can help us adjust our general judgment.